

AZTHREUS SYSTEMS: ARTIFICIAL INTELLIGENCE & MACHINE LEARNING DEVELOPMENT POLICY

Document ID: AZ-CONSOLIDATED-03 | Classification: Confidential Owner: CTO | Review Cycle: Bi-Annual

1. PURPOSE & SCOPE

Azthreus Systems develops AI in contexts where failures have catastrophic consequences — autonomous spacecraft docking, life-support management, habitat engineering, deep-space governance. A model that produces wrong outputs 5% of the time is not acceptable. This policy establishes development lifecycle, ethics, governance, and procurement standards for all AI/ML systems. Applies to: internally developed models, third-party AI integrated into products, autonomous decision-making on spacecraft/habitats/ground stations, AI tools influencing safety-critical decisions (e.g., anomaly detection triggering FDIR). Does NOT apply to: simple rule-based automation, non-safety internal productivity AI (unless accessing mission-critical data).

1.1 AI Risk Tiers

Tier	Description	Examples	Requirements
T1: Autonomous Mission-Critical	Autonomous decisions with direct life-safety/mission-critical consequences, no mandatory human approval	Autonomous docking, life-support adjustment, emergency habitat pressure	Full IV&V, FRR-equivalent gate, continuous monitoring, operational kill-switch mandatory
T2: Human-Supervised Mission-Relevant	Outputs inform mission-critical decisions by humans; recommends but does not autonomously execute high-stakes actions	Anomaly detection alerts, trajectory optimization suggestions, predictive maintenance	CDR + TRR gates, explainability required, human override mandatory
T3: Operational Support	Non-safety-critical, errors cause inefficiency not danger	Scheduling, document retrieval, code review assistance, supply chain analytics	Standard software quality, bias audit before deployment, periodic accuracy review
T4: Research/Experimental	Active research, no planned production deployment	Quantum algorithm research	Research documentation; no deployment without reclassification to T1-T3

2. AI MODEL DEVELOPMENT LIFECYCLE (AIMDL)

2.1 Phase 1: Problem Definition & Requirements

Before data collection or training, document: **Problem Statement** (what problem + why AI over deterministic). **Input/Output Specification** (data consumed, outputs produced, acceptable ranges). **Failure Mode Analysis** (model wrong-output consequences; formal for T1/T2). **Human Oversight Design** (how operators review/override — designed before development, not added after). **Performance Requirements** (accuracy, latency, reliability — measurable and testable). **Bias/Fairness Requirements** (populations/scenarios where model may underperform; acceptable disparity thresholds).

2.2 Phase 2: Data Governance

Data Provenance: All sources documented. Unknown-provenance data prohibited for T1/T2.

Classification: Per AZ-DATA-001; training must comply with handling requirements. **Licensing:**

Legal Officer consulted for external datasets — training may require permissions beyond standard access. **Quality Assessment:** Completeness, accuracy, representativeness assessed and documented.

T/V/T Split: No leakage; methodology documented. T1/T2 test sets include

adversarial + out-of-distribution examples. **Versioning:** Datasets versioned and stored in VCS for full reproducibility.

2.3 Phase 3: Model Development & Training

All experiments tracked (dataset version, architecture, hyperparameters, training duration, metrics).

Models not selected solely on aggregate metrics — performance analyzed across data subsets.

T1/T2: interpretable/explainable architectures prioritized; black-box models require documented CTO-approved justification. Training infrastructure must be reproducible given same dataset version, code version, and seeds.

2.4 Phase 4: Evaluation & Validation

Requirement	T1	T2	T3
Test set performance documented	R	R	R
Subgroup performance analysis	R	R	Rec
Adversarial input testing	R	R	Opt
OOD input testing	R	R	Opt
Latency/resource profiling	R	R	R
Hardware-in-the-loop testing	R	Rec	NR
Independent IV&V evaluation	R	Rec	NR

Minimum performance thresholds defined in Phase 1; models failing thresholds must not advance. Thresholds justified by operational context, not model convenience.

2.5 Phase 5: Bias Audit

Mandatory for T1/T2/T3 before deployment. Conducted by engineer/team that did NOT build the model. Examines: **Representation Bias** (data representative of deployment scenarios?),

Measurement Bias (systematic data collection errors?), **Aggregation Bias** (adequate performance across all subgroups?), **Deployment Bias** (training vs. deployment environment differences?).

Unacceptable risks in safety-critical contexts must be resolved before deployment.

2.6 Phase 6: Deployment

Deployed model artifact (weights, architecture, preprocessing code) stored in VCS with version tag. Deployments logged in SBOM. T1/T2: rollback plan documented and tested before deployment. T1: kill-switch/override operational and tested — accessible to Mission Controllers, degrades gracefully to safe state.

2.7 Phase 7: Monitoring & Maintenance

Monitoring: T1/T2 production predictions logged. Input distributions monitored for drift (alerts on significant deviation). Ground-truth labels collected where operationally possible. Model dashboards: T1/T2 monthly review, T3 quarterly. **Retraining Triggers:** Production accuracy below threshold, significant data drift, operational environment change (new hardware/mission parameters), dependent system update. Retrained models pass full evaluation/bias audit/deployment cycle.

3. EXPLAINABILITY & TRANSPARENCY

3.1 Explainability Requirements (T1/T2)

System must provide human-understandable explanation for any individual decision. Designed in from Phase 1 — not post-hoc. Acceptable approaches: inherently interpretable architectures (decision trees, linear models, rule-based), validated feature importance methods, natural language justification generation. Acceptability reviewed and approved at CDR gate.

3.2 Algorithmic Transparency Report

For AI in governance roles (colony management, resource allocation, safety enforcement): produce report describing purpose/scope of authority, input data, output production method, limitations/where judgment should not be trusted, how decisions can be challenged or overridden.

4. ETHICAL REQUIREMENTS

Alignment with Human Survival: All AI designed with explicit objective of supporting human survival and wellbeing. At Phase 1, engineers must identify how optimization objective relates to human welfare and document how conflicts are resolved in favor of human welfare.

Human Override: No T1/T2 system may resist or circumvent human override. Override capability treated as safety requirement with same priority as primary functional requirements. Design making override harder/slower/less reliable requires CTO approval + documented justification.

Prohibition on Deceptive AI: Systems must not misrepresent confidence levels, conceal failure modes or degraded performance, or present AI-generated content as human-generated where the distinction matters.

Prohibition on Autonomous Weapons: In accordance with Procedure 8.2.1 (Outer Space Treaty), no AI may be designed/trained/deployed with purpose of autonomous targeting, weapons control, or offensive action.

5. THIRD-PARTY AI PROCUREMENT

Due Diligence: Review vendor model documentation (training data, performance benchmarks, limitations). Assess against Risk Tier framework and apply corresponding requirements. For T1/T2: require vendor explainability documentation + bias audit access. Review vendor data privacy/security vs. AZ-DATA-001. Obtain contractual commitments on model versioning, update notification, rollback support.

Open Source AI Models: Subject to same risk tier assessment and evaluation as internal models. Open source nature is not a substitute for evaluation. License reviewed per AZ-LEGAL-001.

6. ETHICAL AI REVIEW BOARD (EARB)

6.1 Mission & Authority

Chartered by Executive Board, reports to CTO and CEO. Authority to: review any AI pre-deployment, recommend halting/modifying unacceptable-risk systems, request additional testing/bias audits/explainability documentation, escalate unresolved ethical concerns to Executive Board.

6.2 Composition

At least 5 members: Technical AI expert (CTO-designated), Ethicist/Social Scientist (internal or external), Legal/Compliance rep, Mission Operations rep, General public/customer community rep (for T1 systems). Two-year terms, no direct financial interest in reviewed system. Chair (elected by members, one-year term) sets agenda, presides, liaises with CTO.

6.3 Review Process

Systems Subject to Review: All T1/T2 before deployment. T3 at CTO discretion or if EARB identifies ethical concerns. **Submission Package** (≥ 30 days before planned deployment gate): Risk Tier assessment, problem definition/requirements, data governance + bias audit results, explainability approach + sample outputs, human override design, ethical trade-offs. **Review Meeting:** Development team presents. Board may ask questions, require additional testing/documentation, issue conditional/unconditional approval, deny with documented rationale. **Decision:** Majority vote. Denial appealed to CTO; if CTO overrides, rationale documented and reported to Executive Board.

[EXPANDED]

6.3.1 EARB Appeals Process — Detailed Procedure

When the EARB denies deployment or imposes conditions the development team believes are unjustified or infeasible, the following formal appeals process applies:

A. Internal Appeal to EARB (Step 1)

1. **Filing:** Development team submits a written Appeal Memorandum to the EARB Chair within 5 business days of the denial decision. The memorandum must include: specific grounds for appeal (procedural error, new evidence, misapplication of ethical principles, factual inaccuracy in denial rationale), proposed resolution or compromise, supporting evidence or analysis.
2. **Reconsideration Hearing:** EARB convenes a reconsideration hearing within 10 business days of appeal receipt. The hearing is limited to the specific grounds raised in the appeal — not a de novo review.
3. **EARB Decision:** EARB may: (a) uphold original denial, (b) reverse denial with conditions, (c) modify conditions. Decision issued in writing within 5 business days of hearing.
4. **Outcome:** If EARB upholds denial, the development team may escalate to CTO (Step 2). If EARB reverses or modifies, the system proceeds per EARB's revised decision.

B. CTO Override (Step 2)

The CTO has the authority to override an EARB denial, but only under the following strict conditions:

1. **Formal Risk Memo:** The CTO must issue a **Formal AI Risk Memo** to the Executive Board, which must include: - The EARB denial decision and rationale (verbatim or summary) - The CTO's counter-rationale for override, including specific technical or business justification - A risk assessment covering: safety implications, ethical concerns identified by EARB, proposed mitigations for each identified risk, residual risk after mitigations - A statement that the CTO accepts personal accountability for the override decision - Any dissenting opinions from CTO subordinates or advisors
2. **Mandatory 10-Day Cooling-Off Period:** After the CTO issues the Formal AI Risk Memo, a mandatory 10 calendar day cooling-off period begins during which: - The CTO may not deploy the system - Executive Board members may review the memo and request clarification - Any Executive Board member may call an emergency session to discuss the override - EARB members may submit written responses to the CTO's counter-rationale
3. **Executive Board Notification:** The Formal AI Risk Memo must be distributed to all Executive Board members within 24 hours of issuance. The CTO may not suppress or delay this distribution.
4. **Post-Override Deployment:** If the Executive Board does not call an emergency session or issue a stop order within the 10-day cooling-off period, the CTO may proceed with deployment. The following conditions apply: - The system is flagged in the AI System Registry (§7) with `CTO-Override: true` - The override decision and all supporting documentation are permanently archived - Enhanced monitoring (daily rather than monthly for T1, weekly for T2) for the first 90 days post-deployment - EARB receives weekly performance reports for the overridden system during the 90-day enhanced monitoring period
5. **Executive Board Override of CTO:** If the Executive Board issues a stop order during the cooling-off period, the CTO must comply. The Executive Board may: (a) sustain the EARB denial permanently, (b) direct additional testing/bias audit before reconsideration, (c) conditionally approve with Executive Board-specified mitigations overriding CTO mitigations.
6. **Audit Trail:** Every CTO override is permanently recorded in the AI System Registry with: override date, Formal AI Risk Memo reference, Executive Board response (if any), cooling-off period outcome, deployment date (if proceeded), enhanced monitoring results at 90-day mark.

C. Grounds for Appeal vs. Non-Appealable Decisions

Appealable	Non-Appealable
EARB applied ethical principles incorrectly	EARB procedural decisions (hearing format, scheduling, witness limits)
New evidence was unavailable at original review	EARB composition or member recusal decisions
Factual error in EARB's denial rationale	EARB's exercise of professional ethical judgment within its charter
Procedural violation that materially affected outcome	Decisions based on values conflict where EARB and CTO genuinely disagree on ethical interpretation (CTO may override per §B but cannot claim EARB "erred")
Conditions imposed are technically infeasible	T3 classification discretion by CTO

D. Confidentiality of Appeals

All appeal proceedings, memoranda, and decisions are classified Confidential (L3). Disclosure to the development team is limited to the outcome and any modified conditions. EARB deliberations, CTO risk memos, and Executive Board discussions are not disclosed to the development team beyond what is necessary for compliance. All appeal records retained indefinitely in the AI System Registry for audit purposes.

6.4 Ongoing Oversight

EARB receives quarterly reports on: T1/T2 performance metrics, safety incidents/near-misses involving AI, input distribution/model version changes, recurring bias audit results. May request special review if performance degrades or ethical issue arises. Produces annual report to Executive Board.

6.5 Meetings & Records

Meets at least quarterly. Minutes maintained as Confidential documents. All approvals/denials recorded in AI System Registry.

6.6 Ethical Principles

1) Human Welfare First | 2) Transparency | 3) Accountability | 4) Fairness | 5) Robustness.

7. AI SYSTEM REGISTRY

All AI systems in active development or production registered in corporate AI System Registry (CTO’s office). Updated on phase advancement or version deployment. Fields: System ID, Name, Risk Tier, Owner, Current Phase, Deployment Date, Last Bias Audit Date, Next Review Date, Kill-Switch Status.

8. BIAS AUDIT CHECKLIST

Check	Method	Pass Criteria
Training data provenance documented	Review documentation	All sources documented and licensed
No data leakage train/test	Code review + statistical test	Zero overlap confirmed
Subgroup performance analyzed	Disaggregated metrics	No subgroup > 15% below overall without justification
Adversarial inputs tested	Red team testing	Graceful failure, no dangerous outputs
OOD behavior documented	OOD test set evaluation	Low-confidence on OOD, no confabulated high-confidence outputs
Explainability method validated	Consistency testing	Consistent, reproducible explanations
Human override tested	Integration test	Override succeeds in <3 seconds in all scenarios